Hearing Drones in 3D: Convolutional Detection and Direction-Finding from a Reduced Microphone Array

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Project Category: Applied — Audio / Convolutional Networks / Defense & Security. **Code:** github.com/danmasad/Acoustic-Drone-Detection

1. Motivation and problem definition

Drones are small, quiet targets that often fly without a radio link, which defeats cameras in poor visibility and radio detectors outright. Sound is a cheap fallback: rotors are loud and harmonically distinctive. From a ground array we want a system reporting, ten times per second, whether a drone is audible and in which direction — azimuth (compass bearing) and elevation (angle above the horizon). The obstacle is hardware: published systems use 24 or more synchronized microphones, and six in one off-the-shelf recorder would cost about four times less. Our reference paper (Rodríguez et al., 2026) names temporal context as its own untested future work, so our question was:

Can temporal context let a cheap 6-microphone array match a 24-microphone one?

The premise turned out to be wrong in an interesting way, which is what this report is about.

2. Related work

Classical localization steers an array and reads off the strongest direction. The canonical method, and our baseline, is **delay-and-sum (DAS) beamforming**: guess a direction, undo the per-microphone delays the geometry implies, sum, and measure energy — a correct guess adds the copies in phase and spikes. Sweeping all candidates turns one frame into an **energy map** over azimuth and elevation whose brightest pixel is the estimate. Refinements (GCC-PHAT, SRP-PHAT, MVDR, MUSIC) reweight the sum; all degrade at low signal-to-noise ratio and share one failure — a **sidelobe**, a direction whose delays align almost as well as the true one, produces a phantom peak one frame cannot distinguish from the source, and thinning an array worsens it.

Learned approaches usually regress direction from multichannel audio. Our reference paper instead treats the map as an *image* and localization as semantic segmentation: a U-Net outputs a mask over the azimuth–elevation grid, and its centroid is the estimate. We adopt this because it makes the array a *preprocessing* choice rather than an architectural one — one architecture trains on any subset, which is what our experiment needs. The paper uses a single frame, evaluates detection only as a stability proxy on five minutes of no-drone audio, and released its dataset and beamforming scripts but no training code, so everything here is reimplemented from its description.

3. Data

The Western Sydney University UAV Acoustic Localization Dataset (CC BY-SA 4.0, 24 GB): open-field recordings of a DJI Air 3 from a 24-microphone array — four independent recorders of six microphones each, 48 kHz — with GPS flight logs at 10 Hz and per-recorder sample offsets. Splits follow the paper: 52 minutes of training across two sessions at site 1, 20 minutes of test 2 at the same site, 3 minutes of test 1 at a different site. **Test 1 was sealed** through all development and model selection. Ground truth comes from the logs, where row k is frame k of the segment — that identity is the whole alignment scheme. Targets are 10°-radius discs; presence is a binary label.

Negative (no-drone) audio is the weak point: the release labels about 5 minutes. We mined the pre-takeoff portion of every recording, which the dataset's own splits never use, verified by beamformed peak contrast that those windows are ambient, and dropped a 20 s margin for motor spin-up. Training negatives went from 4.8 to 8.2 minutes and from one scene to three — still not enough, as section 5.4 shows.

4. Method

Features. One pass over the audio caches every frame's per-microphone spectra (381 bins over 200–4000 Hz, 3.4 GiB), so every downstream choice — array, grid, sub-band count, frame stacking — is a re-steer over the cache rather than another read of 24 GB. Frequency-domain DAS turns spectra into maps of 23 elevations × 91 azimuths at 4°, averaged into 16 sub-bands over 200–2200 Hz: a drone is a harmonic stack, so its true direction lights up in every band and a sidelobe artefact does not.

Model. A depth-3 U-Net (PyTorch), 3×3 convolutions, two heads: a per-pixel mask and a pooled presence logit. Two departures from the reference: circular padding on azimuth, since the map's left and right edges are the same bearing, and the presence head itself, which makes precision–recall measurable. Temporal context is K frames stacked on the channel axis, newest last, so it is causal.

Hyperparameters. Taken from the paper rather than re-searched, since re-searching per condition would confound the comparison: depth 3, Adam at lr 5e-3, class-weighted cross-entropy. Two measured deviations: 32 base filters instead of 64 (2.3× faster *and* better, 72% against 63% within 10°) and a `LeakyReLU` presence head after the plain ReLU died in one seed. Early stopping on the test-2 hit rate, patience 10, best checkpoint restored, identically in every condition — which is what makes the cells comparable. The **baseline** is classical DAS with brightest-pixel decoding, same frames, same metrics.

A separate detector. The presence head reads a beamformed map, so injected external audio would carry a spatial

signature our own rendering invented. We therefore also built a per-channel detector: a 1-D CNN (86k parameters) over one microphone's spectrum, frame decisions by averaging per-channel logits. Exploiting no cross-channel structure, it can eat mono audio unrendered.

5. Experiments and results

All numbers are on test 2 unless stated. Direction is undefined without a drone, so localization is scored on drone frames only, by **angular error** (the great-circle angle between predicted and true direction, folding both axes into one; median, the distribution being heavy-tailed) and **within 10°** (the dataset's own label tolerance). **K** is the number of stacked frames. Headline cells use three seeds with nothing else changed; "±" is the seed sd, "pp" is percentage points. The pooled within-cell seed spread is **±1.2 pp** and is the yardstick throughout — a smaller gap is not a result.

On significance. Three seeds against three cannot return a permutation p below 0.10 however large the effect, so we say that rather than quote a t-test as if it were more. Comparisons *within* one set of weights are far stronger, and there we resample contiguous blocks of frames rather than frames, since a drone crosses a range bin in one pass and treating 11,919 frames as independent would inflate confidence 3.5×. Of the fourteen claims here, corrected together with Holm-Bonferroni, exactly two clear 0.05 — both the history intervention of section 5.3. The rest are point estimates, written as such.

5.1 Does our pipeline reproduce the reference paper?

Their results are published as figures, so comparison meant adopting their conventions: their distance bins, the mean rather than median error (our median against their mean would flatter us by ~16°), and their detection rule, which thresholds no energy but calls a frame detected when the last three direction estimates agree within 10°. Mean angular error for DAS on all 24 microphones over their bins of 0–50, 50–100 and 100–200 m is **38°, 75°, 33° for them against 40°, 72°, 31° for us**, and their stability rule wrongly calls 67.0% of no-drone frames detections against our 68.6%. We reproduce their baseline within a few degrees in every bin, including its non-monotonicity where 100–200 m beats 50–100 m — our strongest end-to-end check, exercising geometry, per-recorder offsets, the STFT, the beamformer, label alignment and the metrics at once against an independent implementation. Our U-Net matches their figures under 100 m under their own rule (7–10° and 9–12° against their 10° and 13°); beyond 100 m it does not, and section 5.5 shows that bin has no acoustic evidence in it.

5.2 Headline: microphone count and temporal context

Rows are microphone subsets: all 24; `device 3`, one off-the-shelf recorder's six microphones, needing no cross-device synchronization; and `best-6`, the six our geometric search called optimal (see below). Only the array and K change.

Drone frames within 10°, mean ± seed sd over three seeds. Last column: what deep context buys.

Array	Mics	Recorders	K=1	K=5	K=9	K=9 − K=1
all 24	24	4	70.5% ± 1.5	70.1% ± 0.9	70.1% ± 2.7	−0.4
device 3	6	1	69.3% ± 0.7	71.1% ± 1.8	**73.2% ± 1.4**	**+3.9**
best-6 (searched)	6	2	52.1% ± 0.8	52.8% ± 1.2	—	—

Six microphones on one recorder match twenty-four: the gap is −1.2 pp at K=1 and +1.0 at K=5, both inside seed noise. Temporal context appears not to help at K=5, and that reading is wrong, for the reason section 5.3 gives. Prompted by it we trained `device 3` at **K=9** — nine frames, 0.9 s — which reaches **73.2% ± 1.4**, +3.9 pp over the same array at K=1, the largest gain anywhere in this table. Because that invited the obvious objection, we then trained all 24 at K=9 as well, and it does not move: 70.1% ± 2.7 against 70.1% at K=5 and 70.5% at K=1.

So **depth pays only where microphones are scarce.** Nine frames of one cheap recorder beat 24 synchronized microphones fed the same nine frames by 3.1 pp — though with three seeds each no cross-array gap is separable (Welch p = 0.18), so the claim that survives is the within-array one: *0.9 s of latency substitutes for 18 microphones, and adds nothing once you already have them.* That is the trade a deployment faces, and the one place our paper's untested "future work" is right.

The third row is a warning about proxies. `best-6` was chosen before any training by scoring all 106,183 non-coplanar 6-microphone subsets on a geometric criterion — the strongest phantom peak more than 15° off the true direction, over 250 simulated directions — and it came *last*, 18 pp behind one off-the-shelf recorder. The criterion is not useless but *over-optimized*: across every array it did not select it predicts accuracy at **−0.80**, and including its own winner collapses that to −0.36. Maximizing `true quality + model error` over 106,183 candidates selects hard on the error term, so the winner of a large search is exactly where a proxy deserves least trust. Cross-recorder synchronization was our first suspect and is dead: two of the three best arrays span two recorders. Notes 22 has all thirteen.

5.3 What temporal context actually buys

A model handed five frames can ignore four of them, so we corrupt the history of *already-trained* K=5 models at test time. `freeze j` keeps the most recent *j* frames real and repeats the oldest kept frame for the rest, holding weights and input shape fixed while **history depth** varies: j=1 has none.

Within 10°, averaged over all nine K=5 runs (3 arrays × 3 seeds). Increments are paired — same weights, two depths — so "±" is the standard error over runs.

History depth	j=1	j=2	j=3	j=4	j=5
Within 10°	60.4%	61.8%	63.2%	64.1%	**64.7%**
Gain (pp)	—	+1.49 ± .21	+1.31 ± .22	+0.97 ± .20	+0.53 ± .07

History is worth +4.3 pp and every step is significant, though the increments are halving. That was the gate for training K=9, and it under-predicted: extrapolating the decelerating increments promised about +1.0 pp for frames six to nine, and the trained cell delivered +2.1. Repeating the intervention on those K=9 weights then found where the climb ends — eight significant increments and a ninth of +0.19 ± 0.11 pp, inside noise. **History saturates at about 0.8 s**, measured rather than assumed, which is what a deployment needs in order to budget its latency. Two further controls (figure 1d): shuffling the five frames costs 0.1 pp and reversing them 0.2 pp, so the model reads the past but is indifferent to its *order* — as the physics demands, since beyond about 70 m the true direction moves less than one grid cell across a whole stack. The usable cue is persistence, which is order-invariant: a drone persists while sidelobes flicker.
This is the one claim here that survives correction. Removing history from the K=9 weights costs **5.1 pp [3.6, 6.8]**, block-resampled, stable from 3 s to 30 s blocks, Holm-adjusted p = 0.0007. It concentrates where one frame is worst: **+0.5 pp inside 30 m, +8.7 at 60–90 m, +21.8 at 90–120 m**, where the hit rate climbs monotonically from 50.8% at one frame to 72.5% at nine. Temporal integration is buying signal-to-noise on faint targets — consistent with the order-invariance, and the same mechanism section 5.5 shows running out of audio past 90 m.
This resolves the apparent null in section 5.2. A K=5 model with its history deleted scores 67.5% on all 24 where a K=1 model trained on single frames scores 70.5%: five times the input channels costs 3.0 pp in optimization and overfitting, and history earns back 2.6 pp. Both are true at once — **the mechanism works, and stacking pays for itself and little more.** Pairing on frame identity shows the same shape across arrays: K=5 − K=1 is negative under 30 m and positive from 60 m out, most strongly where spatial information is scarcest (+8.4 pp at 60–90 m and +10.6 beyond 120 m on `best-6`).

5.4 Detection: the result we had, and why we withdrew it

Detection is scored by **AUROC**: the probability a random drone frame outscores a random no-drone frame, 0.5 being chance and no threshold needed — which matters, since 81% of frames have a drone.
Our first numbers were 0.586–0.727 for the map model's presence head, with a seed spread of 0.07–0.22 and one run at 0.288 (worse than chance), and a stable 0.662 for the per-channel detector. Then we looked at which segments they came from. The training and test ambient are **two halves of one continuous recording** of the same field on the same afternoon, so anything learned about that soundscape transfers to its own second half for free. Our three scenes let the fold boundary be a *scene* boundary instead, each held out in turn, early stopping never touching the held-out one.
AUROC on the held-out scene; chance is 0.5. "Confusers" mixes ESC-50 environmental sound into training frames of both classes, so that "contains an added sound" cannot become the cue.

Held-out scene	Clean	+ ESC-50 confusers
A — different session	0.441	0.347
B — same session as C	0.551	**0.768**
C — same session as B	0.472	0.662
mean ± sd over folds	**0.488 ± 0.056**	0.592 ± 0.219

Cross-scene, detection is at chance — not weak, absent. The 0.662 was measuring the recording, not the drone. The one fold from a *different session* is at or below chance in every variant, while the two folds sharing a session with their training data reach 0.66–0.77: this detector generalizes across flights, not across sites. ESC-50 confusers buy +0.10 on average, entirely in the folds where something already worked.
Our own 5.6 minutes of ambient can only resolve 10.7 false alarms per hour, so we also scored the detector on **DEMAND**: 90 minutes of continuous ambient from 18 environments at our own 48 kHz, level-matched, at thresholds fixed on our own drone frames. At the threshold catching 95% of drones it fires on 60% of DEMAND frames — **21,591 ± 5,532 false alarms per hour** over three seeds, against 93% of our own ambient — and 50% recall still leaves about two per second. No operating point makes this a detector, and the rate is the same on our audio as on eighteen unrelated soundscapes, so it is not an artefact of our field. Consistently, it fires on 76% of ESC-50 sounds mixed into our ambient and on 76% of the eight classes we named as drone-like in advance.
Two fixes that should have worked, and the way they failed is the diagnosis. Both answer obvious complaints: it had never heard real ambient, and it had never been given time. Training on DEMAND itself, holding out whole scene *groups* so the test soundscape is always unheard, cuts false alarms **4.5× on the five groups it trained on and 2.4× on the sixth**. Adding temporal context — the intervention behind our one solid positive result — raises AUROC on our own site monotonically, from **0.691 ± 0.030 at K=1 to 0.752 at K=5 and 0.785 ± 0.053 at K=9**, and rejects at 95% recall from 11% to 19%, while false alarms on unseen DEMAND ambient move the *wrong* way over the same three cells: 21,591 → 27,088 → 28,991 per hour. Neither difference clears correction, and the two

directions together are the point: context sharpens the boundary at the site it was trained on and does not transport it. That is the answer the cross-scene folds and the DEMAND groups already gave, from a third direction. The detector's problem is not capacity, negatives or time — three scenes cannot teach what "not a drone" sounds like anywhere else.

5.5 How far can it hear, and how long does it take to decide?

A pooled AUROC answers neither. Both come out of the same per-frame scores with no retraining: fix a threshold at a false-alarm budget and measure recall per range bin, then average each score over a causal window and watch false alarms trade against latency.

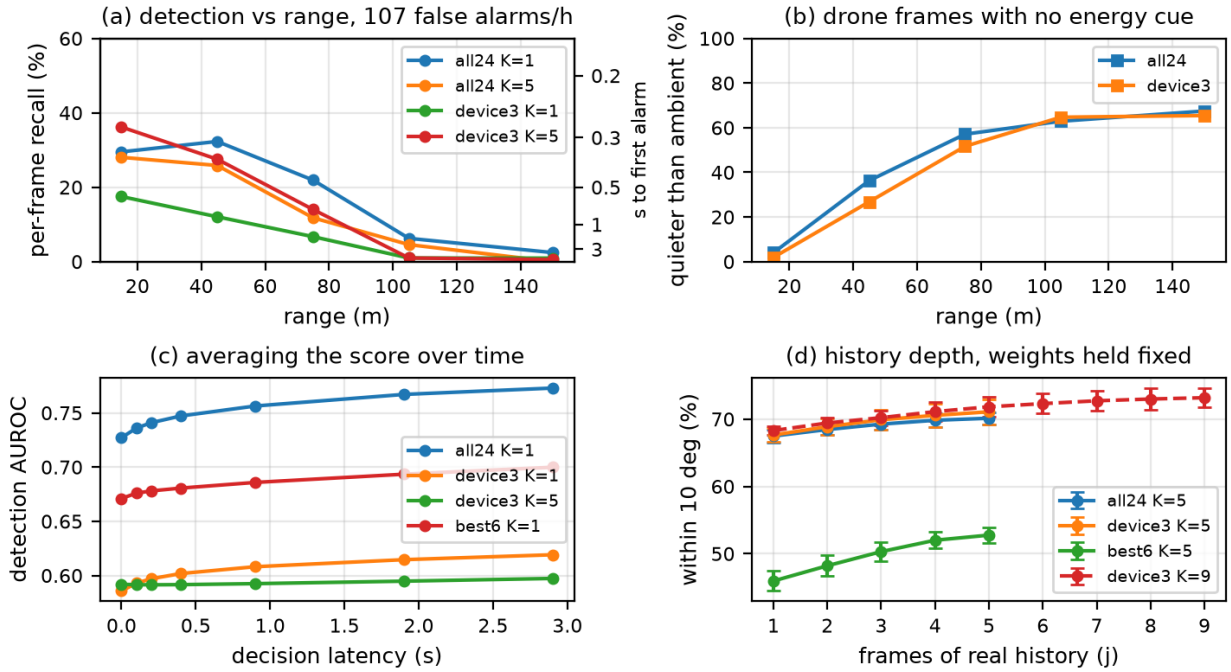


Figure 1: (a) Detection against range at 107 false alarms/hour, the finest rate 5.6 minutes of ambient can express; the right axis converts per-frame recall into the wait for a first alarm. (b) Labelled drone frames quieter than a typical no-drone frame, by beamformed peak energy. (c) Presence scores averaged over a causal window, latency being (window − 1) × 100 ms. (d) Trained models evaluated with 1 to j frames of real history, weights and input width fixed: solid for the nine K=5 runs, dashed for the K=9 cell of section 5.3, where the climb stops.

About 90 m, in under a second. Per-frame recall of 30% sounds like failure and is the wrong unit: frames arrive at 10 Hz and an overflight lasts tens of seconds, so 30% per frame is a first alarm after 0.3 s. Under 90 m every array alarms within about a second; beyond it none alarm at all, and panel (b) says why — two thirds of the labelled positives out there are quieter than typical ambient. Labels come from GPS, not audibility, so past 90 m we score against frames whose strongest map peak is not the drone, which bounds any peak- or mask-decoded model and explains the section 5.1 gap.

Time pays for detection too. A causal average lifts AUROC from 0.727 to 0.773 at 2.9 s of latency, monotonically, in every seed and array. Unlike stacking, this is averaging repeated measurements of a fixed quantity — presence is constant across a window where direction is not. Caveat: no segment straddles a drone's arrival, which is where a deployment would pay the latency.

6. Analysis

Why six microphones suffice. Uniform DAS weights every microphone equally, so a badly placed one actively hurts, and all 24 has two such defects: one recorder is *coplanar*, making directions above and below its plane indistinguishable, and its closest pair sits 0.17 m apart, half a wavelength at 1 kHz, so over most of the band the map repeats (*spatial aliasing*). The learned model recovers most of what uniform weighting discards, after which 18 more microphones buy nothing measurable — and the baseline shows the same ordering, so this is a property of the array, not of the network.

Why temporal context looked useless and is not. Comparing two separately trained models confounds effects of opposite sign: widening the input costs 1.6–6.2 pp before any history is used, and history returns 2.6–6.8. Section 5.2's small number is that *net*, not an absence.

Why detection fails while localization does not. Both are data-limited, in different currencies, and the contrast is the most transferable thing we learned. Localization is supervised by geometry: every drone frame carries a direction from the GPS log, and 52 minutes of flight is tens of thousands of labelled directions. Detection is supervised by *scenes*, and we have three. More ambient frames do not substitute for more soundscapes: 24 channels of one frame are near-duplicates, pooling all of them scoring 0.662 against one microphone's 0.681. Hence an 86k-parameter network at chance across sites while a 2 M-parameter U-Net localizes to 5.5°, and hence a fix that is a microphone and an afternoon

in ten places rather than a bigger model.

7. Limitations and future directions

- **Beyond 90 m the drone is not in the beamformed map.** Two thirds of labelled positives there peak lower than a typical ambient frame, our error is almost all azimuth (53° against 11°), and early stopping on a pooled rate never selected for that bin.
- **The near field is our worst region and we do not know why** — 0–30 m gives 67% within 10° against 78% at 30–60 m, in every seed and array — and section 7.1 shows it is also what fails to transfer. Our steering vectors assume a plane wave, which a source metres from a 2 m array is not: untested.
- **Ablations aimed at known weaknesses returned nulls, and one null was our own bug.** The sub-band count, our reconstruction of a detail the paper omits, moves the hit rate 0.5 pp across 8, 16 and 32 bands; a log-frequency axis for the detector, motivated by a rotor's comb stretching multiplicatively with RPM, gained 0.02–0.06 AUROC against a fold spread of 0.06–0.22.
- **The zenith disc: a null we withdrew, then could not confirm.** Removing the pole distortion on the azimuth axis changed nothing (71.0% ± 1.8) and left the far bin worse — but that version built the disc by *resampling* the finished rectangular map, blurring 4° cells onto 2.5° ones while the target stayed sharp. Steering natively at the disc's own directions costs the same, interpolates nothing, and reverses the sign: mean azimuth beyond 100 m goes from 82.4° resampled and 57.0° rectangular to **48.3° native, 45.9° with the paper's Tversky loss** (which separately costs 6.6 pp of pooled hit rate, so it is a far-field-only trade). Unconfirmed rather than a win: that bin holds two crossings and a ±15° seed spread, so p = 0.43–0.50 — a null is only as good as the implementation under it, and ours was measuring its own interpolation. Nor does it transfer: given `device 3`'s nine frames the disc is 6.2° *worse* there, so the two far-field mechanisms do not compound — both attack a faint peak losing to phantoms, and whichever acts first takes what there is, which is also why 24 microphones and nine frames failed to add. What it buys instead is 2.6° at 50–100 m (9 of 11 crossings, p = 0.065), a lead left open: that puts the fix at high elevation.
- **One drone, one dataset, two sites**, so the cross-airframe question is open and settling it needs an outdoor, uncompressed recording of a comparable airframe. The available alternative — indoor 64 kbps recordings of two Parrot toys — stacks airframe, environment, range and codec mismatches at once, and its no-drone class is ESC-50 itself, which would leak into section 5.4.
- **Untested:** the speed head from the unused GPS velocity labels, the reference paper's skip-attention, and leave-one-flight-out for localization, which is the protocol our detection results say we should have used throughout.

7.1 Test 1: the sealed split, scored once

Three minutes from a different site and date, with its own array calibration, sealed since the first commit and scored by one command after everything above was fixed.

Median angular error in degrees by range over test 1's 1,592 drone frames, three seeds per U-Net cell; one percentage point is 16 frames.

Test 1	within 10°	0–30 m	60–90 m	90–120 m	120–180 m
DAS, 24 mics	**63.9%**	**5.9**	**3.7**	36.3	58.7
U-Net all24 K=1	52.2% ± 5.5	13.3	5.6	5.6	40.5
U-Net dev3 K=5	56.7% ± 7.6	15.1	6.6	**5.3**	27.2
U-Net dev3 K=9	52.0% ± 3.2	20.4	6.5	6.6	**21.2**
share of test-1 frames		*35%*	*30%*	*21%*	*4%*

The headline claim survives. Six microphones on one recorder are not worse than 24 here either: 54.4% ± 4.6 against 52.2% ± 5.5 at K=1 and 56.7% against 53.2% at K=5 — the same direction on both splits, on the cheaper hardware.

Everything degrades and the seed spread quadruples, from ±1.2 pp to ±5–8 pp, so no cell here is separable pooled. The *shape* section 5.3 predicted does survive within `device 3`: K=9 gives its best far error (21.2° against 25.0° at K=1) and its worst near error (20.4° against 14.8°) — deep history paying where evidence is weakest. With 67 frames beyond 120 m, suggestive, not significant.

Classical DAS wins pooled here, and the breakdown says why. Inside 90 m DAS is far better at this site than at ours (5.9° and 3.7°) and still collapses beyond it, while the U-Net is flat at 5–6° out to 120 m. Two thirds of test 1 sits inside 90 m, so the pooled number goes to DAS: what fails to transfer is the learned model's near-field behaviour — already our worst region, and the most geometry-sensitive, this session having its own calibration. **The detector reads 0.639 AUROC**: section 5.4 again on new ground, above chance and worth nothing.

8. Contributions and reproduction

Dan Masad framed the cheap-array question, wrote the pipeline the tables run on (dataset, cache, DAS, U-Net, detector, metrics), ran every experiment those tables quote — including the one sealed test-1 score — and wrote sections 1–7. **Shai Livshits** wrote the four-phase review and notes 43–52: the no-retrain error analyses, then a new baseline under a true test-2 split and leave-one-flight-out. No ablation beat that baseline by more than seed spread, so test 1 was not scored again. Every table regenerates from committed code behind a `GATE PASS` / `GATE FAIL` script; `README.md` has

the commands and `docs/design_notes.md` the full record.

9. References

B. J. Rodríguez et al., *Acoustic Imaging for UAV Detection*, arXiv:2508.00307, IWAENC 2026, with its dataset. Ronneberger et al., *U-Net*, MICCAI 2015. Thiemann et al., *DEMAND*, 2013. Piczak, *ESC-50*, 2015.